

Rate of turn, turn radius and the steering linkage — Northstar 76, 2026 season

26 September 2026. Every number is measured from the boat's own logs. Where the logs cannot support a measurement it is left out rather than estimated, and the coverage is stated beside each figure.

1. The three numbers, and why they differ

Rate of turn — heading change over the **6 seconds centred on the moment the apparent wind crosses the bow**, divided by the span actually measured. The average through the fastest part of the turn.

Turn radius — **speed ÷ yaw rate** over that same window. The radius the boat actually turned at. This is the operating point any steering geometry has to be correct for, and unlike the geometry it can be measured without knowing anything about the boat.

Max rotation — the single fastest heading change between two consecutive samples. A *peak*, which rises with the log's own sampling noise.

Compared **on the same manoeuvres**, max rotation reads higher in **94 of 98 tacks** and **44 of 44 gybes**, by a median of **1.00 °/s** and **1.19 °/s**. That gap is the sampling, not the sailing. A target set from max rotation would be a target nobody is actually sailing.

Timing the window on the apparent wind rather than on the manoeuvre detector matters: on 12 September the crossing lands a consistent **+2.7 s after** the detected moment, so timing from the detector would have measured a different part of every turn.

What the season can carry

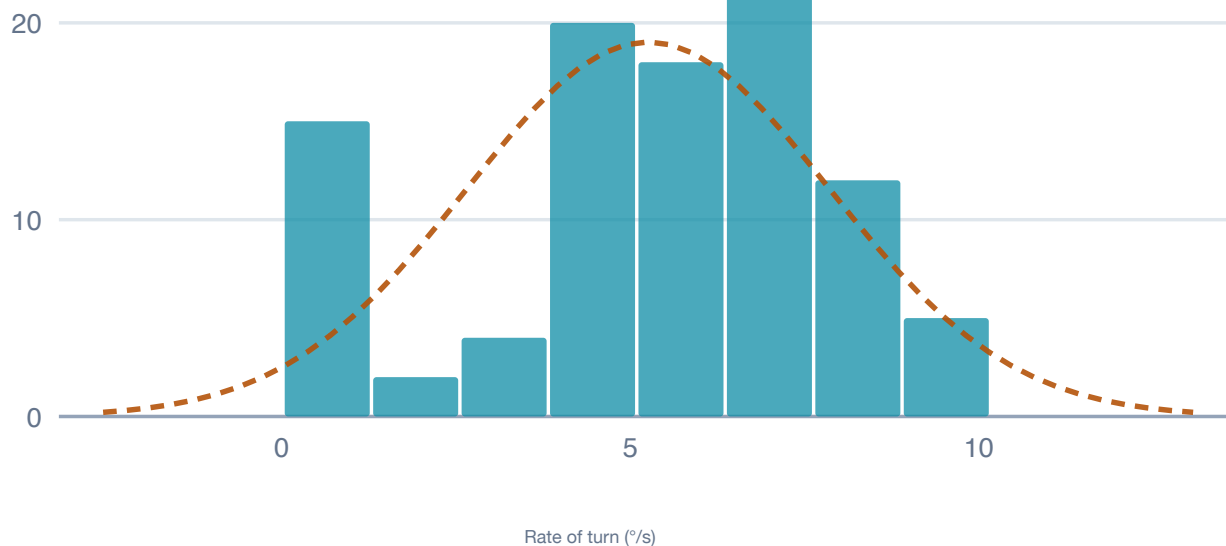
	tacks	gybes
Rate of turn (±3 s window)	98 of 587, on 8 of 23 days	44 of 230, on 6 of 22 days
Turn radius	81	42
Max rotation	546	220

Rate of turn needs a log sampled at **3 s or finer**; half the season is logged at 6.09 s, and a six-second window sampled every six seconds is not a measurement. Turn radius needs the rate *and* a real turn under it — **$R = v/\omega$ runs away as ω goes to zero**, so it is only computed above 2 °/s. Max rotation survives a coarse log, which is why it covers five times as many manoeuvres, and why it should be read as a ceiling rather than a performance.

2. Tacks

Rate of turn across 98 tacks

■ Rate of turn · 98



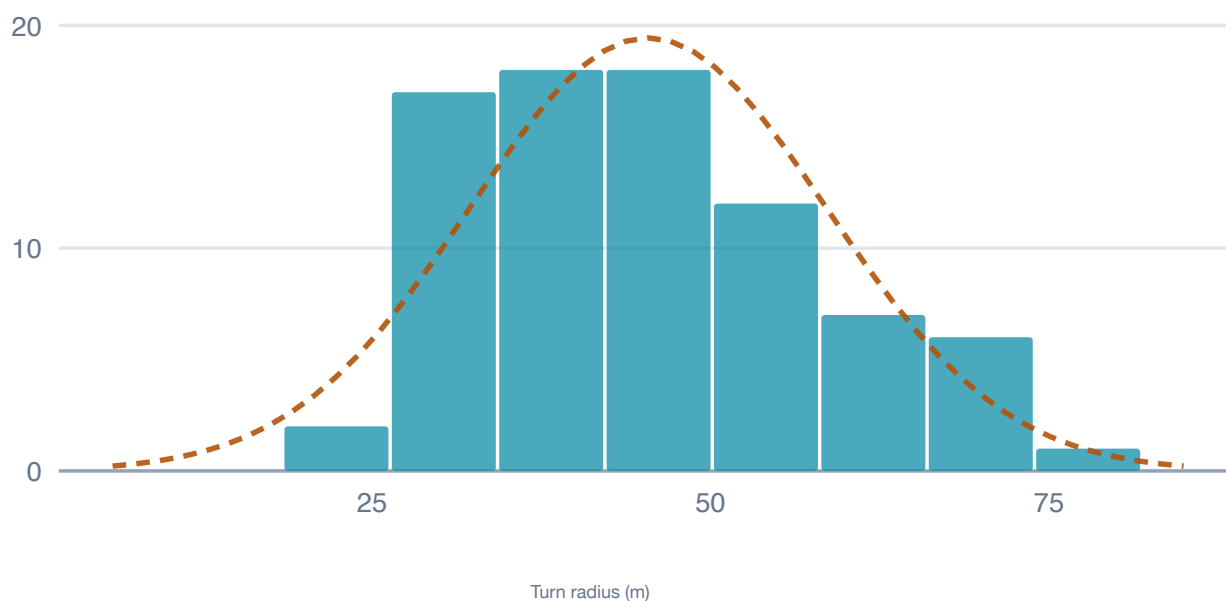
Rate of turn through 98 tacks. Bars are counts; the dashed line is the normal fitted to the same mean and spread, drawn so the difference from it can be seen.

It is not a bell. A hump at 5–7 °/s and a tail of slow ones running to nearly zero. The mean of **5.25 °/s** sits below the median of **5.73** because that tail drags it there; the typical tack is the median.

The bottom bar is not slow tacking. **Fifteen of the ninety-eight** sit near zero, and a tack that turns at 0.3 °/s is something recorded as a tack that was not one, or a stretch where the heading data is bad. They are excluded from the radius for that reason.

Turn radius across 81 tacks

■ Turn radius · 81

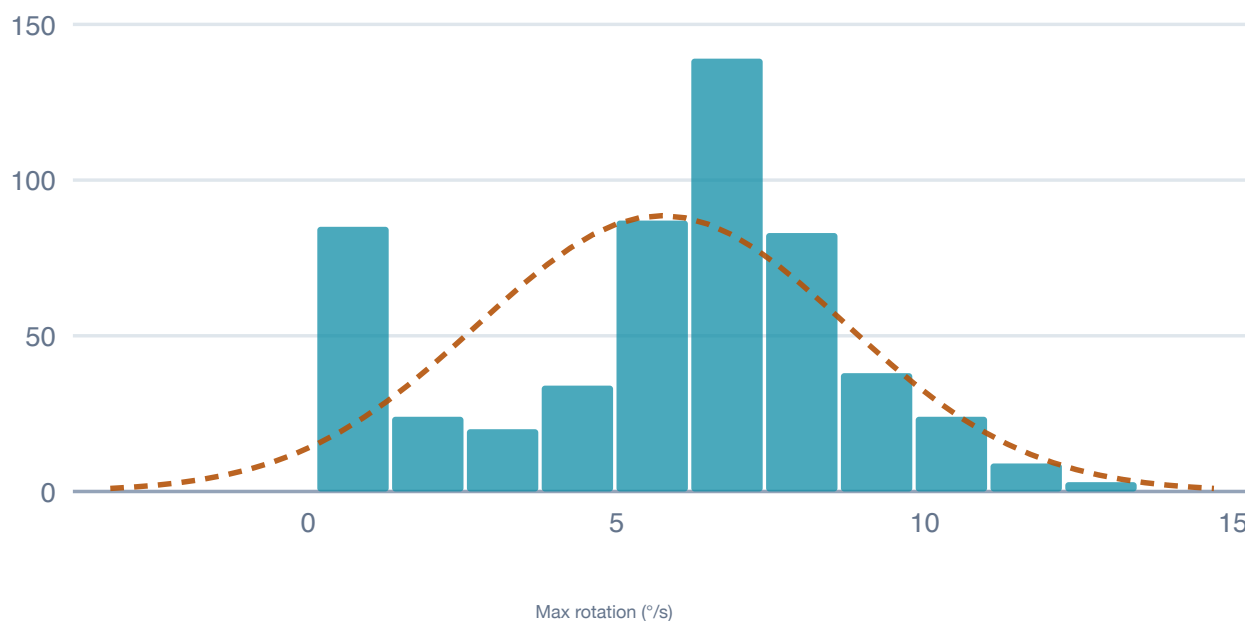


Turn radius through 81 tacks, in metres.

The boat tacks in about 43 m — median, against a hull length of 23.16 m, so **1.86 boat lengths**, and tightly grouped: 18 m to 82 m with a spread of 13 m.

Max rotation across 546 tacks

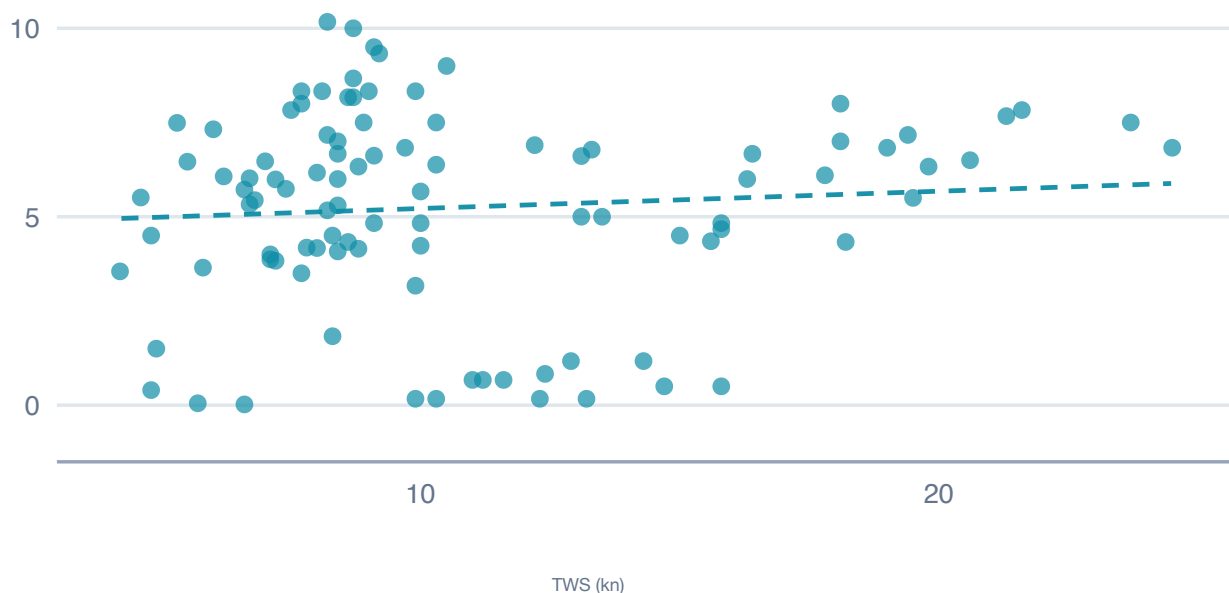
■ Max rotation · 546



Max rotation through 546 tacks — five times the sample, because a coarse log still supports it.

Rate of turn vs TWS (°/s)

● Tacks · 98 · R^2 0.01

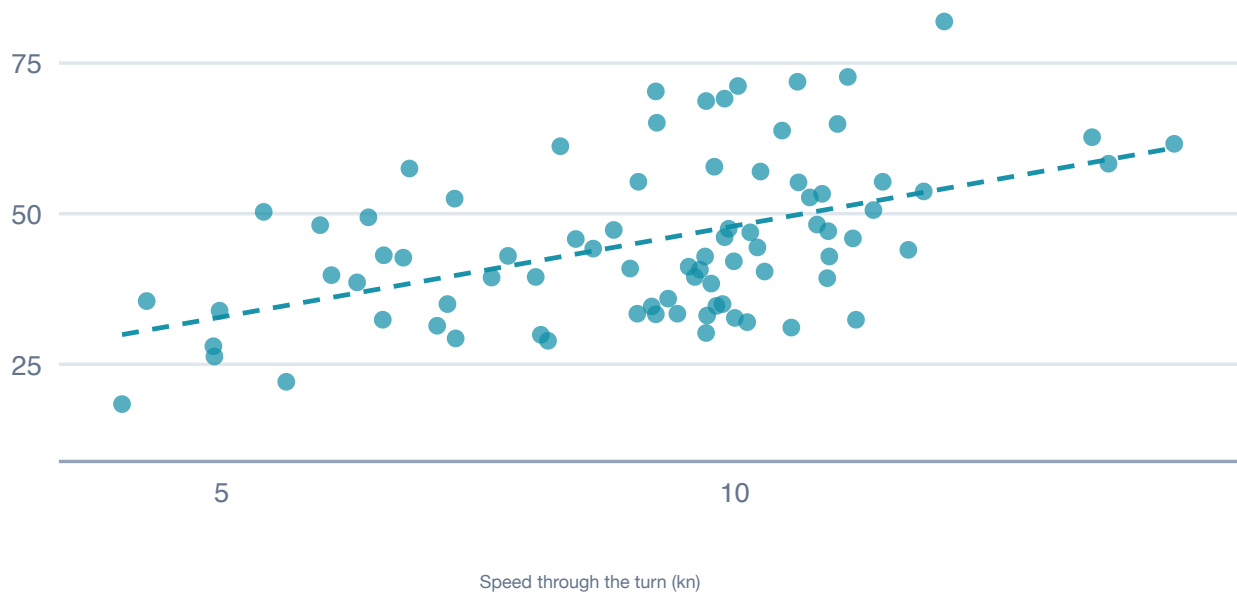


Rate of turn against true wind speed, 98 tacks, with a least-squares line.

Rate of turn does not depend on the breeze. R^2 is **0.01** across 7–20 knots. Whatever governs how fast this boat comes through the wind, it is not wind strength — so it is worth looking at what the crew does rather than at conditions.

Turn radius vs Speed through the turn (m)

● Tacks · 81 · R^2 0.24

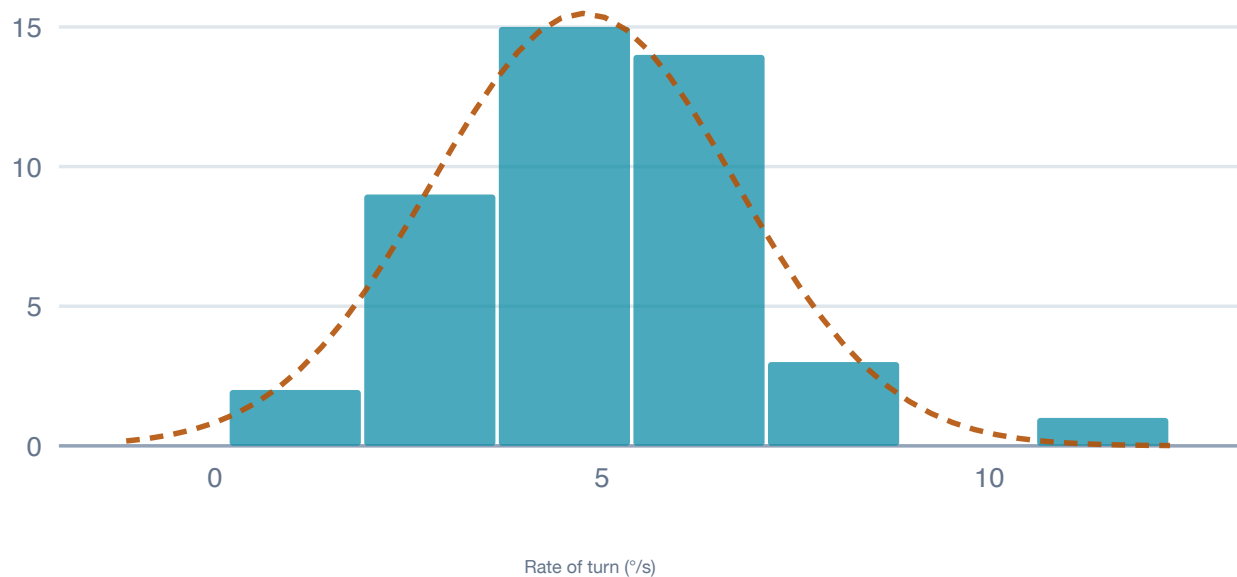


Turn radius against the speed carried through the turn.

3. Gybes

Rate of turn across 44 gybes

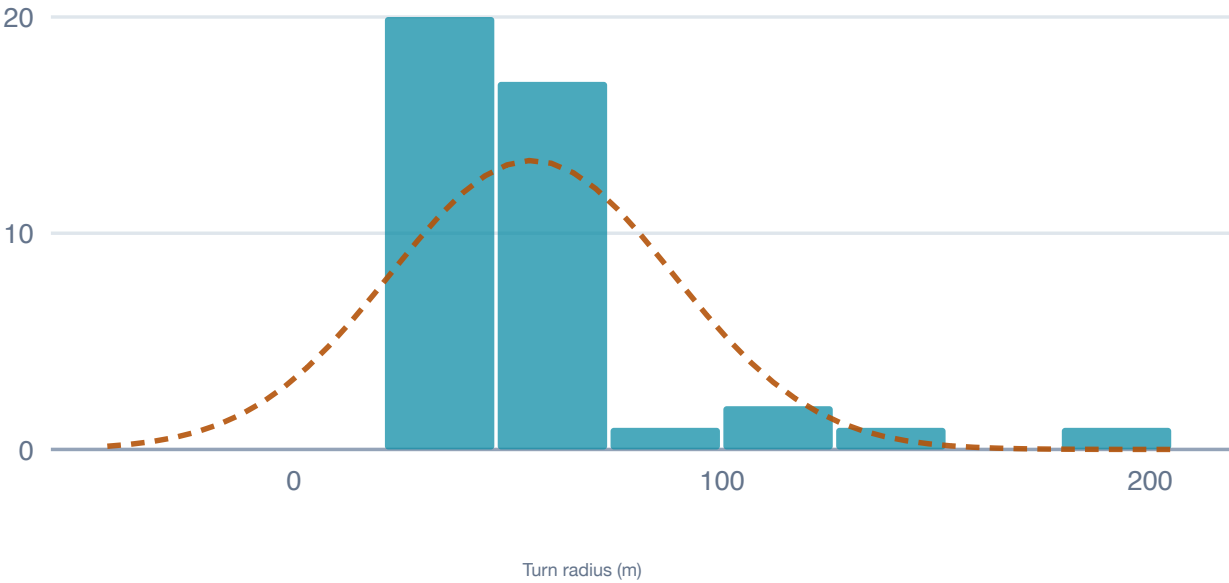
■ Rate of turn · 44



Rate of turn through 44 gybes.

Turn radius across 42 gybes

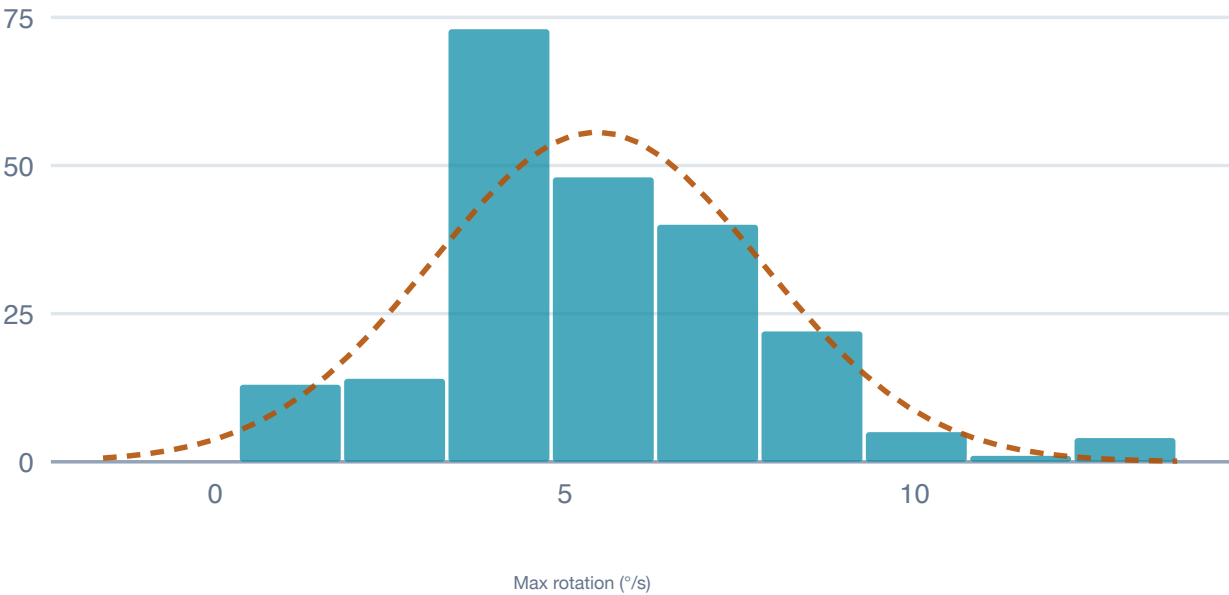
■ Turn radius · 42



Turn radius through 42 gybes.

Max rotation across 220 gybes

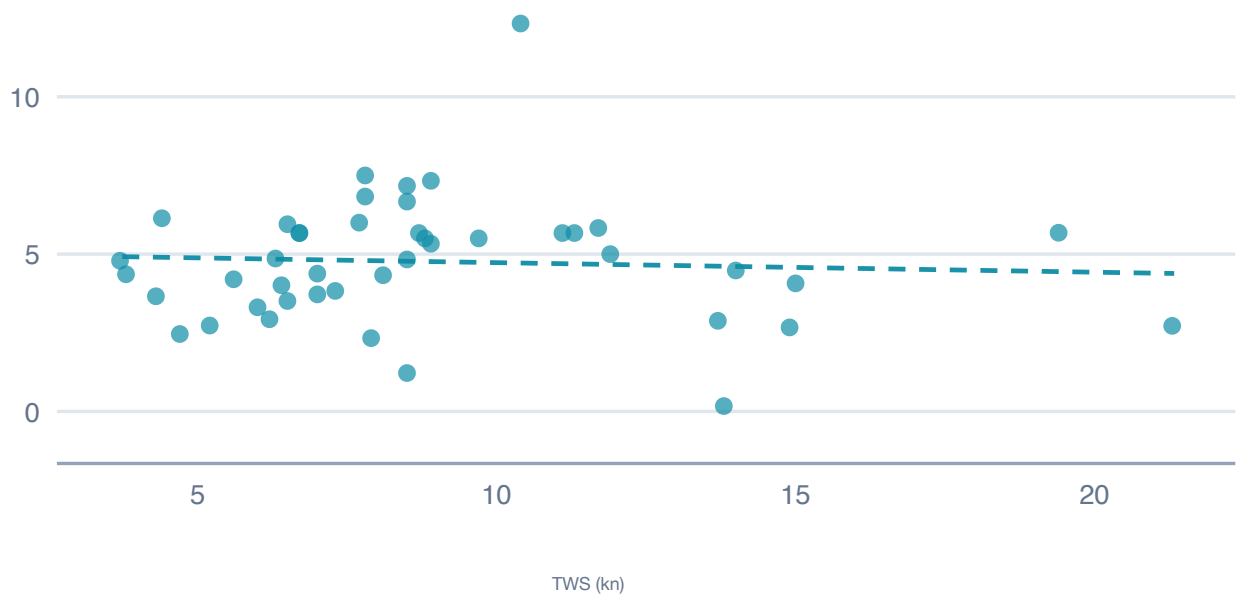
■ Max rotation · 220



Max rotation through 220 gybes.

Rate of turn vs TWS (°/s)

● Gybes · 44 · R² 0.00



Rate of turn against true wind speed, 44 gybes.

Gybes turn **slower and wider** than tacks — 4.81 °/s against 5.73, and a radius of **49.6 m (2.15 boat lengths)** against 43.0 — but the rate is more tightly grouped (spread 1.97 against 2.61). A gybe is a smaller heading change made under more control, and it is the more repeatable manoeuvre on this boat. The radius spread is the wider of the two (33 m against 13 m), driven by a handful of very wide ones out to 205 m.

4. Summary

- The typical tack turns at **5.7 °/s** in **43 m**; the typical gybe at **4.8 °/s** in **50 m**. Medians, from the ± 3 s window.
- The boat turns in **roughly two boat lengths**, and tacks are the tighter of the two manoeuvres.
- The tack rate distribution is **not normal** — a hump with a tail of slow ones. The tail is where the time goes, not the average.
- **Rate of turn is independent of wind speed** (R^2 0.01). If it is to be improved, the variable is not the breeze.
- **Fifteen of the ninety-eight rated tacks look like bad records rather than bad tacks** and should be checked before any mean is quoted onward.
- Max rotation reads higher on the same manoeuvre almost every time. Use it for coverage, not as a target.

5. Steering geometry

The operating point, measured

This is what turn radius buys: the geometry only has to be right where the boat actually sails, and that is now known rather than assumed.

Treating the boat as a bicycle — a single effective steering angle at an effective distance **L** ahead of the turn centre — the helm carried through each turn gives

$$L = R \cdot \tan \delta$$

Across **68 tacks** that lands at a median **12.6 m** (range 5.5–19.6), and across **36 gybes** at **11.7 m**. The two agree, which they need not have, and both are close to **half the hull length**.

That is an *effective* pivot distance, not a tape measurement: a hull generates side force along its whole length, so it will not equal the geometric distance from the rudder stock to the centre of lateral resistance. It is, however, the number the boat behaves as though it has — **worth comparing against the tape measurement rather than replaced by it**.

At the measured operating point — radius **43 m**, L **12.6 m** — the helm angle implied is **16.3°**, which matches the top of the measured steering range.

The linkage, measured

Across **78,149 log rows** carrying both rudder angles, fitting starboard against port gives

$$ruddS = 1.0008 \times ruddP - 1.215 \quad (R^2 \ 0.932)$$

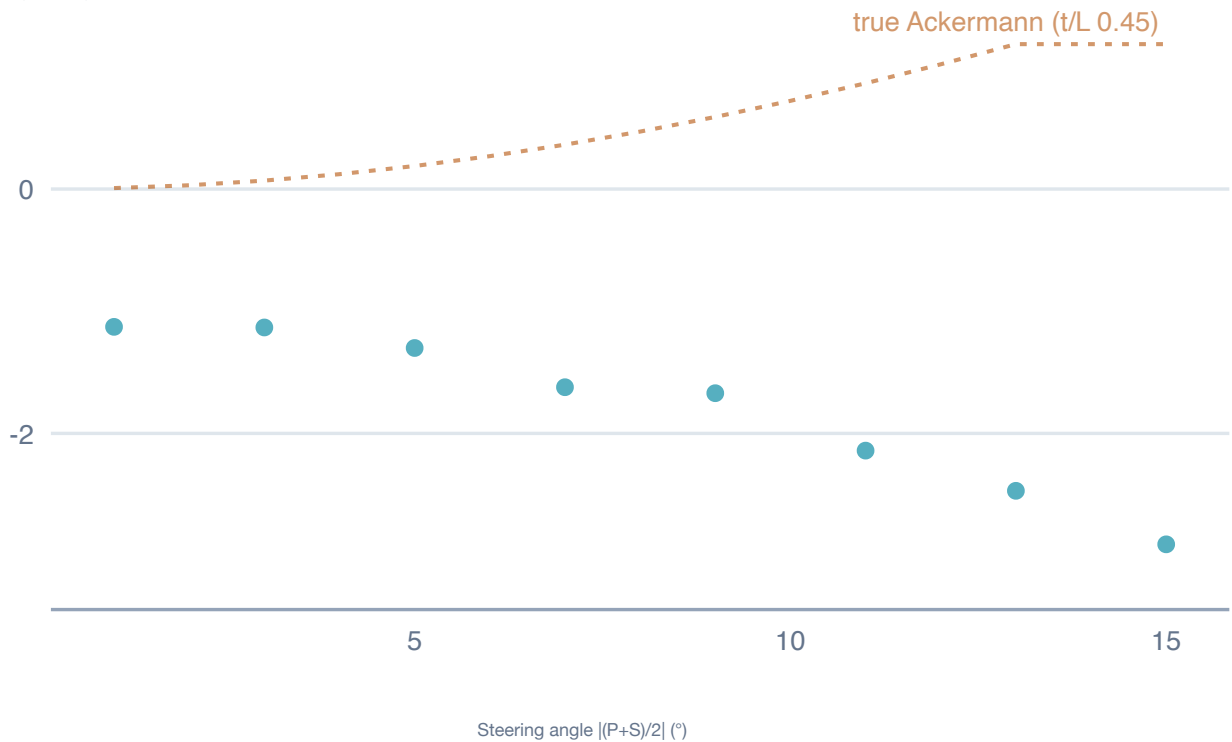
A slope of **1.0008** is **parallel steering**: within these sensors' resolution the two rudders turn by the same amount, with a **constant offset of about –1.2°** rather than a progressive differential.

The differential does drift with steering angle — weakly, and the *wrong way*:

steering angle	rows	mean starboard – port
0–2°	43,058	-1.13°
2–4°	19,228	-1.13°
4–6°	6,996	-1.30°
6–8°	3,633	-1.62°
8–10°	1,990	-1.67°
10–12°	1,359	-2.14°
12–14°	1,042	-2.47°
14–16°	842	-2.91°

Rudder differential against steering angle (°)

● Measured (binned) · 78149



Measured differential (binned means) against steering angle, with the differential true Ackermann would require shown dashed. The measured curve runs the other way.

About **-0.09 to -0.15° per degree** of helm depending on turn direction, with R^2 of only **0.05 to 0.14** — a small systematic effect inside a lot of scatter.

Under true Ackermann the inner rudder turns **more** than the outer, making starboard – port **positive** in both directions. It is negative and grows more negative. **The boat is running slightly anti-Ackermann.**

The target, once the rudder separation is known

With **L = 12.6 m measured** and the operating radius of **43 m**, the only missing number is **t**, the separation between the rudder stocks. Inner and outer angles at that operating point:

t (m)	t / L	inner	outer	differential needed
4.0	0.317	17.08°	15.64°	+1.44°
4.5	0.357	17.18°	15.56°	+1.62°
5.0	0.397	17.28°	15.48°	+1.80°
5.5	0.437	17.38°	15.40°	+1.98°
6.0	0.476	17.48°	15.32°	+2.17°

Against a **measured -2.91°** at that helm. Whatever **t** turns out to be, the correction is of the order of **4.4° to 5.1° at full helm** — reversing the sign and adding the Ackermann. That is large enough to be worth getting right, and large enough that guessing **t** would have been irresponsible.

The one thing that must happen first

These are sensor readings, not linkage geometry. A constant **-1.2°** offset is indistinguishable, from the log alone, from one rudder's angle sensor being zeroed **1.2°** differently from the other. **Zero both sensors dockside against physically aligned rudders and record the offsets before anything mechanical is touched.** If the offset proves to be calibration, what remains is the **~0.1 °/°** drift, which is small — and the whole picture changes.

The linkage is not a tie rod — and that changes the answer

The first version of this section solved a **steering trapezoid**: one tie rod running directly between the two tillers. That is the wrong mechanism, and its answer (a ball joint 105 mm inboard on the tiller) does not apply. **Disregard it.**

What the boat actually has is **two rods to a central slider**: each tiller is joined by its own rod to a single slider on the centreline, low in the hull, and the slider travels athwartships. The Ackermann in that arrangement comes from somewhere else entirely — not from the tiller's angle, but from **how far fore-and-aft the slider sits from the rudder stocks**.

It is also, usefully, a much better mechanism for this: it can be made to track true Ackermann almost exactly, where the trapezoid could not.

The geometry as built, from the CAD

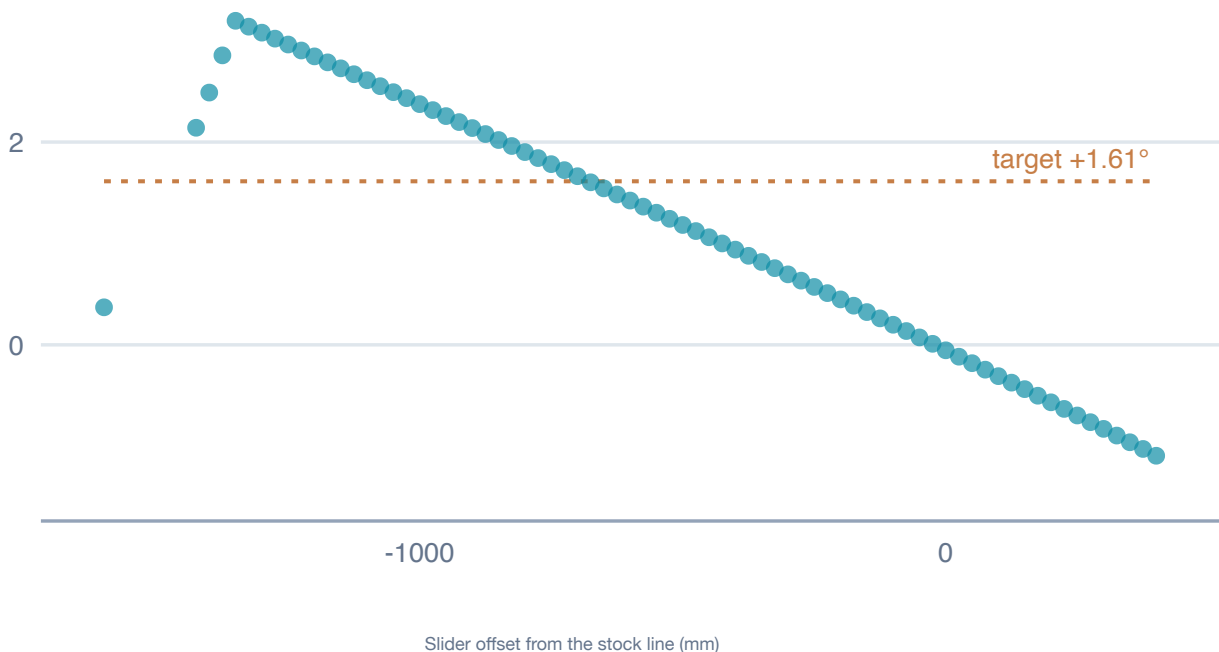
Stock axis at tiller height	x 22382.83, y 1727.84, z 548.31
Tiller ball joint	x 22041.45, y 1665.90, z 502.19
Tiller vector	−341.4, −61.9, −46.1 → 350.0 mm
In plan	346.95 mm , angled 10.3° inboard of the fore-and-aft line
Out of plane	46.1 mm down, 7.6° below horizontal

Two separations, and they are not interchangeable. At the tiller the stocks are $2 \times 1727.8 = \mathbf{3456\text{ mm}}$ apart — that is the number the *linkage* works with. The $2 \times 2238 = \mathbf{4476\text{ mm}}$ quoted earlier is presumably at the blade, and that is the number the *Ackermann target* is set by, because that is where the rudders make their side force. The 1020 mm difference is consistent with a raked stock, and using either one for both jobs would give the wrong answer. Both are used below, each in its place.

The slider position is the control

Ackermann against slider fore-and-aft position (°)

● Delivered · 75



What the linkage delivers as the slider is moved fore and aft, measured at the 15.6° of helm the boat carries in a 43 m turn. Negative is toward the tiller ends.

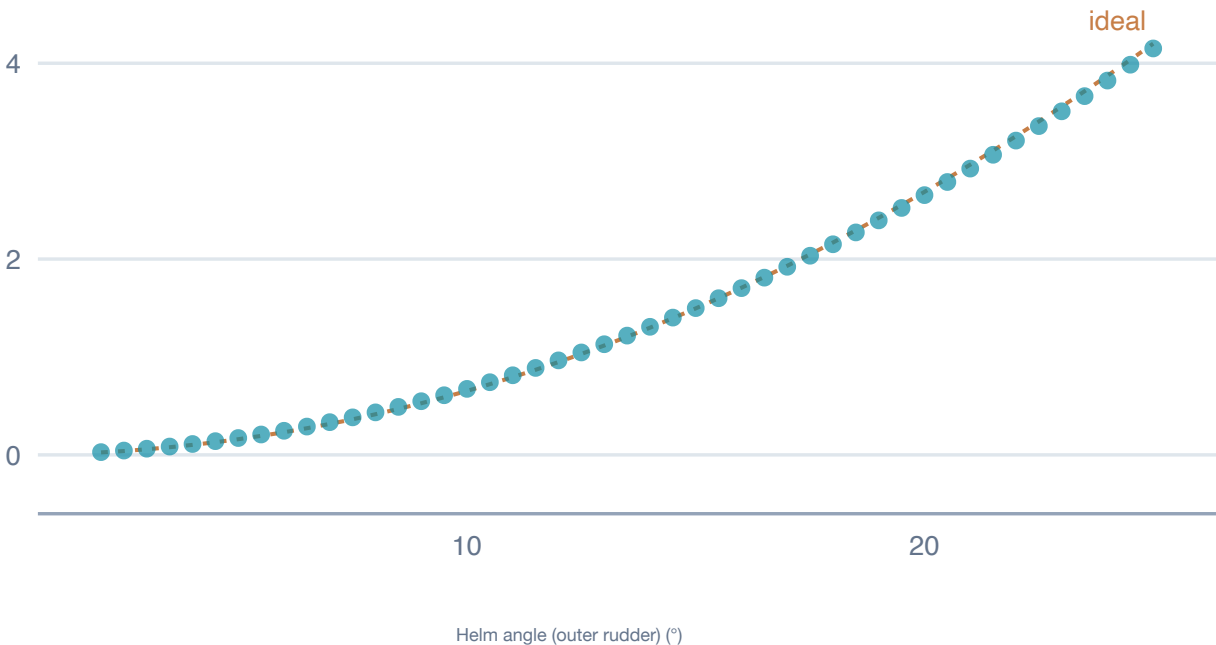
The differential swings from **+3.6° to -4.4°** across the slider's plausible range, passing through zero when the slider is level with the stocks. There is therefore exactly one position that gives the +1.61° the measured turn radius calls for:

Slider ball joint	CAD x = 21704 mm, y = 0
relative to the stock line	679 mm toward the tiller ends
relative to the tiller ends	338 mm beyond them, same direction
Rod length in plan	1761 mm
True rod length	$\sqrt{(1761^2 + \text{drop}^2)}$ — set by how far the slider sits below the tillers

The rod's vertical drop does **not** affect the steering at all: the tiller sweeps a circle in one plane and the slider travels in another, so the vertical separation is constant and only the horizontal projection does any work. The drop sets the rod's true length and nothing else.

With the slider at -679 mm: delivered vs ideal (°)

● Delivered by the linkage · 47



With the slider at that position, delivered against ideal across the whole helm range.

helm	delivered	ideal	error
5°	+0.17°	+0.16°	+0.01°
10°	+0.67°	+0.65°	+0.02°
15.6°	+1.61°	+1.61°	0.00°
20°	+2.65°	+2.69°	-0.03°
25°	+4.15°	+4.20°	-0.05°

Within **0.05° everywhere from 5° to 25°** — it does not just hit the design point, it follows the curve. That is the two-rod slider earning its keep.

How precisely it has to be placed

slider offset from ideal	differential	off target
-200 mm	+2.09°	+0.48°

slider offset from ideal	differential	off target
–100 mm	+1.85°	+0.24°
–50 mm	+1.73°	+0.12°
0	+1.61°	0.00°
+50 mm	+1.49°	–0.12°
+100 mm	+1.37°	–0.24°
+200 mm	+1.13°	–0.48°

±50 mm is worth getting right; ±10 mm is not worth chasing.

Do the two rudders match what the turns required?

The most direct test available, and it does not depend on any model of the linkage: take the rudder angles actually recorded through each manoeuvre, and compare them with what that manoeuvre's own measured radius required.

Tacks — 72 manoeuvres, median radius 42.1 m:

	now	required	change
inner rudder	15.31°	17.54°	+2.24°
outer rudder	18.26°	15.86°	–2.39°
mean of the two	16.78°	16.70°	–0.08°
differential	–2.95°	+1.68°	+4.63°

Gybes — 38 manoeuvres, median radius 46.4 m: measured differential **–1.75°** against a required **+1.40°**, an error of **–3.15°**.

Two separate findings, and they point in opposite directions.

The pair is right. The mean of the two rudders — 16.78° — is within a tenth of a degree of the 16.70° that a 42 m turn asks for. The boat is carrying the correct amount of helm. (Some of that agreement is built in: L was derived from $R \cdot \tan \delta$ in the first place, so the mean *had* to come out close. It is the differential that is an independent measurement.)

The split between them is wrong, and by more than the correction is. The outer rudder is turning **3 degrees more** than the inner, where it should be turning 1.7 degrees **less**. The inner rudder is under-turned by **4.6° on a tack** and **3.2° on a gybe** — and since the rudders sum correctly, every degree the inner is short is a degree the outer is carrying in excess, dragging.

That also corrects something said earlier in this document

An earlier draft concluded that no plausible linkage could produce the measured anti-Ackermann — that arms would need splaying 28° outboard — and inferred the drift was therefore probably sensor error. **That was the wrong mechanism being modelled.** Against the two-rod slider the boat actually has, the measurement is entirely ordinary: a slider at **679 mm the wrong side of the stocks** produces exactly what was recorded.

Working backwards from the measured **–2.95°** at 16.8° of helm, the slider that reproduces it sits at **stock_x +1091 mm — CAD x ≈ 23474 mm** — on the *opposite* side of the rudder stocks from the tiller ends.

That is a checkable prediction, and it is the next thing to look at. If the slider really is about 1.1 m the far side of the stock line, the geometry explains the whole measurement, nothing is wrong with the sensors' gain, and the fix is purely mechanical:

Move the slider from $x \approx 23474$ to $x = 21704$ — about 1770 mm, to the other side of the rudder stocks.

If it is somewhere else, the model is wrong somewhere and that is worth knowing before anything is moved.

A 1770 mm move is a large one and may not be practical as drawn. The same differential can be reached by changing the tiller geometry instead — the arm angle and the ball joint position — and once the slider's real position is known the alternatives can be costed against each other. What is not in doubt is the size of the error: **4.6° of differential at the radius this boat actually tacks at.**

What is still needed

1. **The slider's position as built** — CAD x, y, z of its ball joint, and its travel axis. Without it there is no way to say how far it needs to move, only where it should end up. This is the one number that turns this into a work order.
2. **Confirm 2238 mm is the blade separation**, not something else. The Ackermann target scales with it.
3. **L, the pivot distance.** Still the 12.6 m the boat behaves as though it has. From 9 m to 16 m the slider position moves by roughly ± 250 mm, so this is worth the tape measurement.
4. **The stock rake.** The tiller arm sits 7.6° below horizontal, which is what a tiller perpendicular to a raked stock looks like. The model above treats the tiller as sweeping a horizontal circle; with real rake it sweeps a tilted one, which is a second-order correction to the numbers above — small, but worth redoing once the rake angle is known.

And still, before anything is cut

The **-1.2° constant offset** — the differential still present at zero helm — is the part that remains indistinguishable from one rudder sensor being zeroed differently from the other. The *drift* with helm angle is now fully explained by the linkage, but a constant offset at zero helm is not, and it is worth an hour to settle: **zero both sensors dockside against physically aligned rudders.** It also decides whether the -2.95° above should be read as -2.95° or -1.8° .

In order

1. **Zero both rudder sensors** dockside. This alone may account for the -1.2° .
2. **Read off the slider ball joint's x, y, z** and its travel axis from the CAD.
3. **Confirm the blade separation and measure L.**
4. **Then** move the slider to **x = 21704 mm** on the centreline — adjusted for whatever L turns out to be — and set the rod length to suit its height.
5. **Re-measure on a 1–2 s log.** Plotted against the second chart above, the differential either crosses to positive and follows the ideal, or it does not.

Generated from `session_phase_stats` and the session logs, 26 September 2026. Linkage solved numerically as a symmetric steering trapezoid (two tiller arms, one tie rod, rudder separation 4.0 m), verified against the parallel-arm case which must and does return exactly parallel steering. Rate of turn: heading change over the 6 s centred on apparent wind = 0, on logs of 3 s or finer, gated on the crossing being interpolated across ≤ 3 s and the window holding ≥ 3 headings. Turn radius: mean boat speed over the same window divided by the yaw rate, above $2^\circ/\text{s}$ only. Linkage fit: ordinary least squares over log rows carrying both rudder angles with $|\text{steering}| \leq 15^\circ$. Effective pivot distance: $R \cdot \tan \delta$ with δ the mean helm over the turn, 2° – 25° .